

Demo 4 — Rice GWAS

$p \approx 158,000$ **SNPs**, $n = 1,155$ **rice accessions**, a **known grain-length locus**

Table of contents

The biology	1
The data	1
Running COMBSS (sketch)	2
R	2
Python	2
The selected SNPs	2
Best-subset inclusion path	3
Correlation among selected SNPs	4
Why this matters	4
Reference	5

This demo uses pre-computed results — the raw genotype data is too large to ship with this site. The code shows what `combss` was called with; the figures below were produced by that call.

The biology

Grain length is one of the most economically important traits in rice breeding. The major effect locus has been mapped to the **GS3 gene on chromosome 3**, identified independently by multiple GWAS groups using thousands of accessions.

Can COMBSS recover this locus *de novo* from raw SNP data, working at the scale of a real GWAS?

The data

- $n = 1,155$ rice accessions (3K Rice Genomes Project subset)
- $p = 158,210$ SNPs after QC (MAF filter, call-rate filter)
- Binary response: long-grain vs short-grain (threshold at 6 mm)
- Population stratification corrected via the top principal components of the kinship matrix

Source: McCouch et al. (2016), *3,000 rice genomes project*. Data preparation and QC are described in Mathur, Liqueet, Muller, Moka (2026).

Running COMBSS (sketch)

Once `dataYX` has columns `[y, PCs..., SNP1, SNP2, ..., SNP_p]`:

R

```
library(combss)

y      <- dataYX$y
x_snps <- as.matrix(dataYX[, -c(1, 2)]) # drop y and intercept-like column

fit_rice <- combss(x_snps, y,
                  family = "binomial",
                  q = 10)
fit_rice$subset_list
```

Python

```
from combss import combss

y_rice = dataYX["y"].to_numpy()
X_snps = dataYX.drop(columns=["y", "intercept"]).to_numpy()

fit_rice = combss(X_snps, y_rice, family="binomial", q=10)
print(fit_rice.subset_list)
```

The call returns a nested-looking subset path for $k = 1, \dots, 10$. We tabulate the selected SNP indices below.

The selected SNPs

The model at each size k :

	k	Selected
k1	1	43318
k2	2	43307, 43318
k3	3	43307, 43313, 43318
k4	4	43307, 43313, 43318, 66479
k5	5	43307, 43309, 43313, 43318, 66479
k6	6	43307, 43309, 43313, 43318, 66479, 66480
k7	7	43304, 43307, 43313, 43318, 43359, 66479, 66480
k8	8	43307, 43309, 43313, 43318, 43335, 66479, 66480, 157373
k9	9	43307, 43313, 43318, 43335, 43336, 66479, 66480, 157368, 157373
k10	10	43288, 43307, 43313, 43318, 43335, 43336, 66479, 66480, 157368, 157373

The single best SNP at $k = 1$ — index **43318** — falls inside the **GS3 gene region on chromosome 3**, the well-known grain-length locus. SNPs **43307**, **43313** (chromosome 3 neighbours of GS3) enter at $k = 2, 3$. The cluster around **66479**, **66480** (chromosome 5) and **157368**, **157373** (chromosome 12) join the model as k grows, picking up secondary loci known from independent GWAS.

Best-subset inclusion path

The full inclusion-path figure (Figure 4a in the paper):

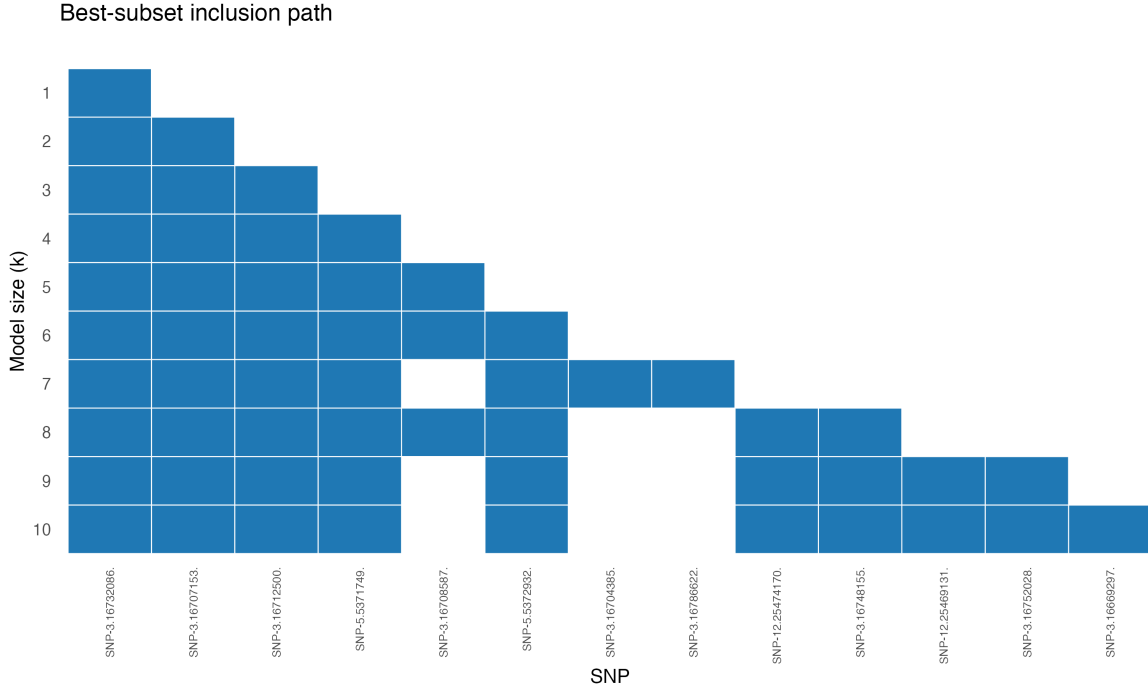


Figure 1: Best-subset inclusion path for the rice GWAS. Each column is one selected SNP; each row a model size k . A blue tile means the SNP is selected at that k . SNPs are ordered by first-inclusion.

The earliest-included SNPs are tightly clustered on chromosome 3 — the GS3 locus. Additional chromosomes contribute as k grows.

Correlation among selected SNPs

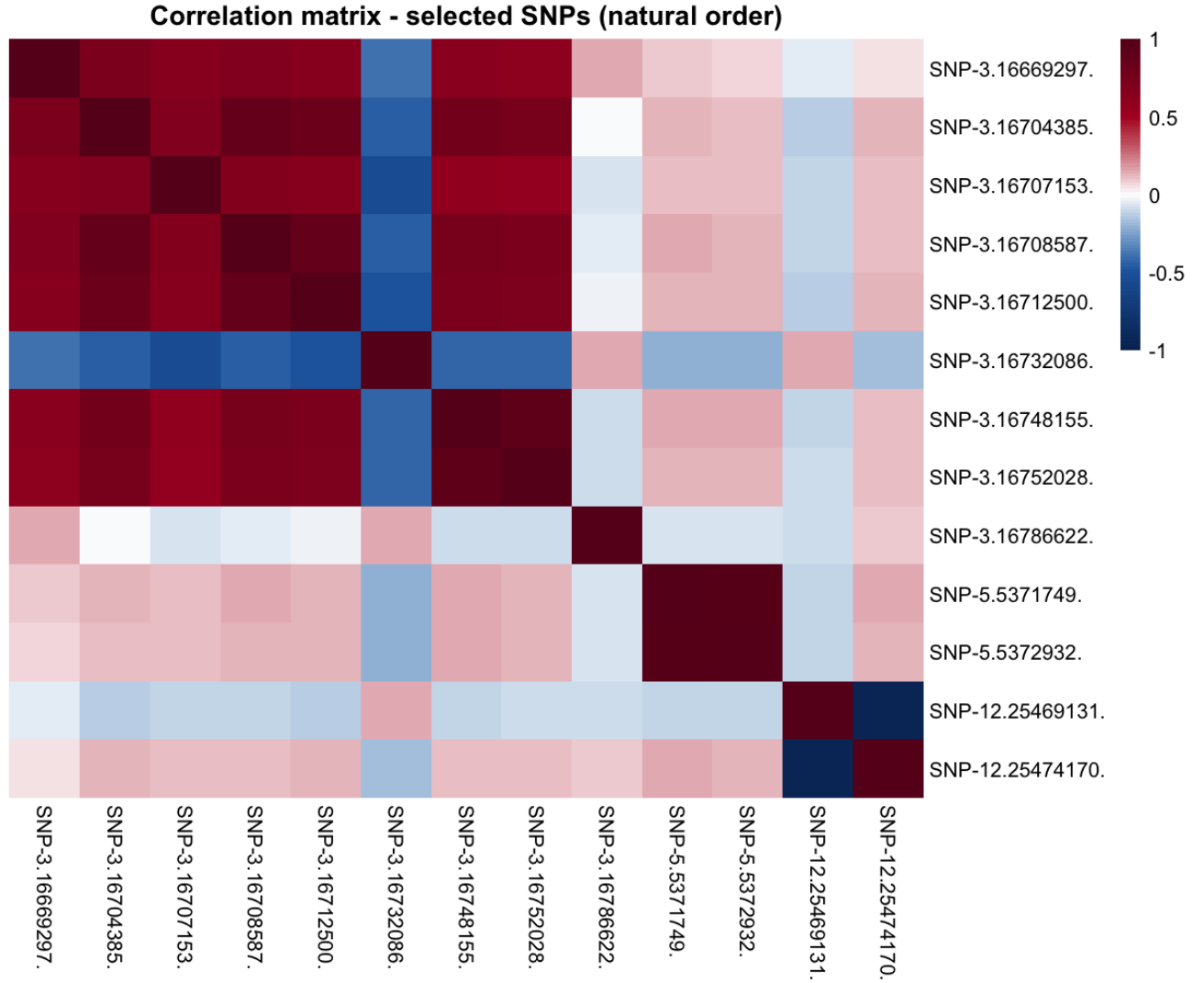


Figure 2: Correlation matrix of the SNPs selected across all model sizes. Strong off-diagonal blocks reveal linkage disequilibrium within each chromosomal cluster.

The block-diagonal structure tells the right biological story: SNPs from the same chromosomal region are in strong linkage disequilibrium, while across chromosomes they are nearly uncorrelated.

Why this matters

- COMBSS scans 158{,}210 SNPs in a few minutes on a laptop — the regime where exact best-subset is hopeless and lasso typically returns hundreds of correlated SNPs.
- The single best SNP it returns lands on the **known** functional locus, with no prior biological input.
- For a breeder, *ten SNPs* is a tractable downstream panel; *several hundred* is not.

Reference

McCouch, S. R. et al. (2016). *Open access resources for genome-wide association mapping in rice*. **Nature Communications** 7, 10532.